

- 5 (a) People with Alzheimer's disease (AD) lose their ability to form new memories. One form of Alzheimer's disease, called familial Alzheimer's disease, is caused by an autosomal dominant allele of the *APP* gene.

To study Alzheimer's disease, identical genetically modified mice containing the dominant human *APP* allele have been produced. These mice are known as AD mice and are used as mouse models of Alzheimer's disease.

When these AD mice are trained to swim through a water maze, they perform poorly and cannot learn as well as normal mice.

- (i) Suggest what steps will be needed to make identical genetically modified AD mice.

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.....[4]

- (ii) Suggest why it is useful to have an animal model of a human disease.

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- (b)** Researchers wanted to know if changes in gene expression were important in the inability of the AD mice to learn.

Groups of normal mice and AD mice either received training to allow them to learn how to swim a water maze, or they received no training. The mice in the four groups then had mRNA extracted from the memory-forming areas of their brains.

Reverse transcription of the mRNA of individuals in each group was carried out and the resulting cDNA was labelled with fluorescent nucleotides. This was then used for DNA microarray analysis using slides containing DNA sequences from **33 696** mouse genes.

Explain the principles of this type of DNA microarray analysis.

[4]

(c) Table 5.1 summarises the microarray analysis of differences in gene expression for:

- an untrained AD mouse compared to an untrained normal mouse
- a trained AD mouse compared to a trained normal mouse.

Table 5.1

training received	number of genes expressed in AD mouse but not in normal mouse	number of genes expressed in normal mouse but not in AD mouse	total number of genes showing a difference in expression between the normal mouse and AD mouse
no	17	11	28
yes	112	820	932

- (i) Calculate the percentage of mouse genes whose expression has been shown to be affected by training.

Show your working.

percentage = % [2]

- (ii) State what the results in Table 5.1 show about the effect of training and learning on gene expression in brain cells.

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[1]

- (d) The genes which are expressed in the brains of normal mice undergoing training and learning code for proteins important in synapse and memory formation. A large number of these genes are under the control of one transcription factor, a protein called *Crtc1*.

To try to improve learning in AD mice, researchers caused over-expression of the *Crtc1* gene in the brains of AD mice, by delivering the gene to mouse brain cells using a virus vector.

- (i) State the name given to this type of treatment.

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- (ii) Suggest the effects of over-expression in the brain of the *Crtc1* gene on AD mice.

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[Total: 15]